



[BIT703 Web Technologies](#) / [Contents](#) / Topic 7: Client-side web application development and programming – JavaScript / Basics of JavaScript

Introduction to JavaScript 15m

This section will provide you with a brief overview of the basics of JavaScript. You will see that you have no required tasks for this section, however, we have provided links for further readings should you require more knowledge.

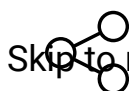
What is JavaScript used for?

JavaScript is a scripting language that enables you to create interactive and dynamic content, control multimedia and animate images.

JavaScript runs inside client web browsers, where the script code is embedded in, or called from, HTML documents. These scripts can generate page content dynamically. JavaScript code can be used to manipulate data, user interfaces and graphical elements. JavaScript can be used server-side through platform tools such as node.js.

In general, JavaScript can be used to:

- improve the visual look and feel of the user's experience of a web application, such as pop-up windows and interactive menus
- offload some of the web application processing from the server to the client, such as performing client-side form validation – checking the contents of an HTML form filled in by the user before it is submitted to the server for further processing
- enable asynchronous JavaScript and XML (often referred to as Ajax) implementations that provide a more seamless interaction between client and server.

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If you need a refresher about this, we recommend taking a look at the following

Client-side technologies →
JavaScript frameworks →

The video *What is JavaScript and why do I need it?* (Ask Leo!, 2023) (8:37) explains the fundamentals of JavaScript and its significance in modern web development.



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What is JavaScript and Why Do I Need It?

What Is JavaScript and Why Do I Need It?

Ask Leo!



Watch on



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Features of JavaScript

JavaScript is the most widely used programming language in web development.

Here are some facts about JavaScript and its features:

- JavaScript was first created for Document Object Model (DOM) manipulation. Earlier websites were mostly static, but after JavaScript was created, websites became dynamic.
- Functions in JavaScript are objects. They may have properties and methods just like other objects. They can be passed as arguments in other functions.
- JavaScript can handle dates and times.
- JavaScript performs form validation, although the forms are created using HTML.

Advantages and disadvantages of JavaScript

Like all computer languages, JavaScript has certain advantages and disadvantages. Many of these are related to it being a client-side language.

Advantages	Disadvantages
Speed. Being client-side, JavaScript is very fast. Any code functions can be run immediately instead of having to contact the server and wait for an answer.	Security. Because the code is executed on the user's computer, in some cases it can be exploited for malicious purposes. This is one reason some people choose to disable JavaScript.
Simplicity. JavaScript is relatively simple to learn and implement.	Reliance on the client. JavaScript is sometimes interpreted differently by different browsers. Whereas server-side scripts will always produce the same output, client-side scripts can be a little unpredictable. Don't be overly concerned by this, though. If you test your script in all the major browsers, you should be safe.

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Versatility. JavaScript plays nicely with other languages and can be used in a huge variety of applications. Unlike PHP and other scripts, JavaScript code can be inserted into any web page or included as an independent file. JavaScript can also be used inside scripts written in other languages, such as Perl and PHP.	Disabled JavaScript. If JavaScript is disabled in the browser, then the JavaScript will not run.
Server load. Being client-side reduces the demand on the website server.	File download. A JavaScript file is downloaded on a client machine so anyone can read the code and reuse it (exceptions exist when the code may be obfuscated to make it intentionally harder to read).
	Local access. JavaScript does not interact with files/resources outside of the browser space (except when JavaScript is used in purpose-built tools like VS Code).



Thoughts on this page?

